

Qubits & Superposition — Study Notes

A concise, visual introduction to the qubit, superposition, measurement, and practical context — designed for undergraduate learners.

1. Why Qubits Matter

Classical bits encode a single deterministic value (0 or 1). Qubits encode quantum information and can exist in linear combinations of 0 and 1 simultaneously. That extra flexibility is the foundation for quantum speedups and new algorithms.

- **Fundamental unit:** A qubit is the basic unit of quantum information.
- **Non-classical behavior:** Superposition and entanglement enable phenomena with no classical analogue.
- **Algorithmic potential:** Qubits underlie algorithms like Shor's and Grover's that outperform classical approaches for specific problems.

2. What Is a Qubit?

Computational basis states are represented in column vector notation:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These form an orthonormal basis for a single qubit system.

General state of a qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex probability amplitudes. The amplitudes encode the likelihoods of outcomes when measured.

■ **Key Concept:** Think of α and β as directional weights — both magnitude and phase matter in quantum behavior.

3. Normalization & Probabilities

The complex amplitudes α and β satisfy the fundamental normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

Measurement in the computational basis yields a classical result with probabilities:

- $P(0) = |\alpha|^2$
- $P(1) = |\beta|^2$

After measurement, the qubit collapses to the observed basis state and loses its prior superposition.

4. Superposition — Intuition & Example

Superposition means the qubit simultaneously encodes components of $|0\rangle$ and $|1\rangle$ until observed.

Example: Equal Superposition State

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \rightarrow \text{Equal probability of measuring 0 or 1 (50\% each).}$$

- **Key Implication:** Quantum parallelism allows a single qubit (and multi-qubit registers) to represent many possibilities at once.

- **Limit:** Measurement yields only one outcome per run. Algorithms use quantum interference across amplitudes to amplify desired results.

5. Bloch Sphere — Geometric View

The Bloch sphere represents any single-qubit pure state as a point on a 3D unit sphere. Coordinates (θ, ϕ) map to:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

Phases (ϕ) are physically important for interference but invisible to measurement probabilities alone.

6. Physical Implementations & Applications

Common physical qubit platforms include **superconducting circuits**, **trapped ions**, **photons**, **quantum dots**, **neutral atoms**, and **nuclear spins**. Each platform trades off coherence time, control precision, and scalability.

Key application domains: **Quantum Cryptography & Secure Communication**, **Drug Discovery & Molecular Simulation**, **Optimization (logistics, finance)**, and **Quantum Machine Learning**.